

InvoiceAI

Smart Invoicing & AR Automation for Indian SMBs

System Design Document

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1. Executive Summary

InvoiceAI is an AI-powered invoicing and accounts receivable (AR) automation platform purpose-built for Indian small and medium businesses. The platform addresses the critical pain point of late payments and manual collection efforts that drain SMB resources.

The system generates GST-compliant invoices across all categories (B2B, B2C, and export), tracks payments in real time, and deploys intelligent automated collection reminders through WhatsApp and email with AI-drafted messages that escalate in tone from polite to firm to final notice.

Key differentiators include:

- WhatsApp-native AR follow-up leveraging India's dominant business messaging platform
- AI-powered payment prediction that identifies which clients are likely to pay late
- Automatic bank statement reconciliation to match payments with invoices
- Deep integration with India's GST infrastructure including e-way bill generation
- Payment link generation through Razorpay, PayU, and UPI for frictionless collection

The platform targets the 63 million MSMEs in India, many of which still manage invoicing through Excel or paper. InvoiceAI aims to reduce Days Sales Outstanding (DSO) by 40% through automated, intelligent AR management.

2. System Architecture

InvoiceAI follows a modular, event-driven architecture designed for reliability, scalability, and ease of integration with India's payment and tax infrastructure.

2.1 Technology Stack

Layer	Technology
Backend API	FastAPI (Python 3.12) with async support
Frontend Dashboard	React / Next.js 14 with TypeScript
Primary Database	PostgreSQL 16 with row-level security
Cache & Queue	Redis 7 (BullMQ for job processing)
AI / ML	OpenRouter (free-tier models) for AI drafting
Messaging	WhatsApp Business API (via Gupshup/Twilio)
Payments	Razorpay, PayU, UPI deep links
File Storage	S3-compatible (MinIO / AWS S3)
PDF Generation	WeasyPrint / Puppeteer for invoice PDFs

2.2 Architecture Overview

The system comprises five major subsystems that communicate through typed domain events:

- **Invoice Engine:** Generates, manages, and stores GST-compliant invoices with automatic tax calculation based on place of supply rules (CGST+SGST for intra-state, IGST for inter-state).
- **Payment Tracker:** Monitors payment status, processes webhook notifications from payment gateways, and reconciles bank statement imports against outstanding invoices.
- **AR Automation Engine:** Manages reminder sequences with configurable escalation timelines and AI-drafted message content that adapts tone based on client payment history.
- **Communication Layer:** Abstracts WhatsApp Business API and email (SMTP/SES) delivery, handles template management, and tracks delivery/read receipts.
- **Analytics & Reporting:** Generates aging reports, cash flow forecasts, client payment scoring, and GST filing reports (GSTR-1, GSTR-3B).

2.3 Key Design Decisions

- **Event-driven architecture** using BullMQ ensures reliable reminder delivery even under high load or temporary failures.
- **Multi-tenant with PostgreSQL RLS:** Every table includes `business_id`. Row-level security policies enforce tenant isolation at the database level.
- **Idempotent payment processing:** Payment webhooks are deduplicated using transaction IDs to prevent double-processing.
- **Offline-resilient:** Invoice PDFs are pre-generated and cached, allowing WhatsApp delivery even if the generation service is temporarily unavailable.

3. Data Model

The database schema is designed around multi-tenant isolation with `business_id` as the primary partition key across all tables. All monetary values are stored in `paise` (integer) to avoid floating-point precision issues.

3.1 Core Tables

Table	Purpose	Key Fields
businesses	Tenant organizations	gstn, pan, address, plan_tier
clients	Business customers/debtors	gstn, email, phone, payment_score
invoices	Invoice headers	invoice_no, type, status, due_date, total_paise
line_items	Invoice line items	description, hsn_code, qty, rate_paise, tax_rate
payments	Payment records	amount_paise, method, gateway_txn_id, status
payment_links	Generated payment URLs	provider, short_url, amount_paise, expires_at
reminders	Individual reminder records	channel, tone, sent_at, delivered_at, read_at
reminder_sequences	Reminder escalation templates	name, steps (JSONB), active
bank_transactions	Imported bank statement entries	date, amount_paise, description, matched_invoice_id
recurring_invoices	Recurring invoice schedules	frequency, next_date, template_invoice_id
credit_notes	Credit notes and refunds	original_invoice_id, reason, amount_paise
tax_settings	Business GST configuration	gstn, state_code, composition_scheme, hsn_defaults

3.2 Key Relationships

- invoices belong to businesses and clients; each invoice has many line_items
- payments reference invoices (many-to-one for partial payments)
- reminders belong to invoices and follow a reminder_sequence template
- bank_transactions can be matched to payments via reconciliation
- credit_notes reference the original invoice they adjust
- recurring_invoices generate new invoices on schedule based on a template

4. Core Features

4.1 GST-Compliant Invoice Generation

The invoice engine supports all Indian GST invoice types:

- B2B invoices with GSTIN validation and automatic CGST/SGST or IGST calculation based on place of supply
- B2C invoices for consumer transactions under the B2C threshold
- Export invoices with LUT/bond support, zero-rated GST, and multi-currency amounts
- Debit and credit notes for adjustments and returns

Each invoice automatically generates a PDF using configurable templates branded with the business logo, terms, and bank details. HSN/SAC code lookup is built in with automatic tax rate determination.

4.2 Recurring Invoices

Businesses can configure recurring invoice schedules (weekly, monthly, quarterly, annually) with automatic generation and delivery. The system creates new invoices from templates, adjusts dates and numbering, and sends them via the configured channel (email, WhatsApp, or both).

4.3 AI Payment Prediction

The payment prediction model analyzes historical payment patterns to score clients on their likelihood of paying on time:

Score	Category	Avg Days Late	Action
90-100	Reliable	0-3 days	Standard reminder sequence
70-89	Moderate Risk	4-15 days	Proactive early reminders
50-69	High Risk	16-45 days	Aggressive follow-up + payment links
< 50	Chronic Late	> 45 days	Require advance payment or COD

4.4 WhatsApp AR Automation

The AR automation engine sends payment reminders through WhatsApp using pre-approved message templates. Messages are drafted by AI and escalate in tone based on the number of days past due.

4.5 Bank Statement Reconciliation

Businesses can import bank statements (CSV, OFX, or PDF) for automatic reconciliation. The matching engine uses a combination of amount matching, date proximity, reference number extraction, and client name fuzzy matching to link bank transactions to outstanding invoices. Unmatched transactions are flagged for manual review.

4.6 Payment Link Generation

InvoiceAI generates one-click payment links through Razorpay, PayU, and UPI that are embedded directly in invoice PDFs and WhatsApp reminders. Links are short-lived (configurable expiry) and support partial payments.

4.7 Additional Features

- Credit note and refund management with automatic GST adjustment
- Multi-currency support for export invoices with real-time exchange rates
- Client payment history and scoring dashboard
- Aging reports (current, 30, 60, 90, 120+ days) with drill-down
- Cash flow forecasting based on invoice pipeline and payment predictions
- E-way bill integration for goods movement above threshold

5. AR Automation Flow

The accounts receivable automation follows a configurable escalation sequence. Below is the default flow:

Day	Channel	Tone	Sender	Content
Day 0	WhatsApp + Email	Informational	System	Invoice sent with payment link
Day -1	WhatsApp	Friendly	System	Payment due tomorrow reminder
Day +1	WhatsApp	Polite	System	Gentle reminder with payment link
Day +7	Email	Firm	System	Formal follow-up with statement
Day +15	WhatsApp + Email	Final Notice	System	Final payment notice with consequences
Day +30	Email	Escalation	Business Owner	Personal escalation to business owner

Each reminder includes the invoice PDF, outstanding amount, and a one-click payment link. The AI drafts message content using the client's payment history and communication preferences. Business owners can preview and approve AI-drafted messages before they are sent, or enable auto-send for reliable clients.

The escalation sequence is fully configurable. Businesses can adjust timing, add or remove steps, customize tone guidelines, and set different sequences for different client tiers.

6. API Design

InvoiceAI exposes a RESTful API with JWT authentication and API key support for integrations. All endpoints are versioned under `/api/v1/`.

6.1 Invoice Endpoints

Method	Endpoint	Description
POST	<code>/invoices</code>	Create a new invoice
GET	<code>/invoices</code>	List invoices with filtering and pagination
GET	<code>/invoices/:id</code>	Get invoice details
POST	<code>/invoices/:id/send</code>	Send invoice via WhatsApp/email
GET	<code>/invoices/:id/pdf</code>	Download invoice PDF
POST	<code>/invoices/:id/payment-link</code>	Generate payment link

6.2 Payment & Client Endpoints

Method	Endpoint	Description
POST	<code>/payments/webhook</code>	Receive payment gateway webhooks

POST	/payments/reconcile	Upload bank statement for reconciliation
GET	/clients/:id/score	Get client payment score and history
GET	/reports/aging	Generate aging report
GET	/reports/cashflow	Cash flow forecast

7. Revenue Model

InvoiceAI follows a freemium SaaS model designed to convert SMBs from free to paid as their invoicing volume grows:

Plan	Price	Features
Free	Rs. 0/mo	Up to 10 invoices/month, basic email reminders, 1 user
Professional	Rs. 999/mo	Unlimited invoices, WhatsApp reminders, bank reconciliation, 3 users
Business	Rs. 2,999/mo	AR automation, AI payment prediction, recurring invoices, API access, 10 users
Enterprise	Custom	Custom integrations, dedicated support, SLA, unlimited users

8. Integration

InvoiceAI integrates with the broader Apprend Technologies product ecosystem and third-party services:

8.1 Internal Integrations

- **GSTBot:** Two-way sync for GST filing. InvoiceAI pushes invoice data to GSTBot for GSTR-1 filing, and GSTBot validates GSTIN and pulls tax rates.
- **GoSumo:** Client management integration. New clients created in either system are synced. InvoiceAI can send reminders through GoSumo's WhatsApp channel.

8.2 External Integrations

- **Tally Export:** Generate Tally-compatible XML for businesses using Tally for accounting
- **Razorpay & PayU:** Payment link generation, webhook processing, settlement reconciliation
- **UPI:** Deep link generation for direct UPI payments
- **WhatsApp Business API:** Template message delivery via Gupshup or Meta Cloud API
- **GSTN:** E-invoice registration and IRN generation for B2B invoices above threshold
- **E-way Bill:** Automated e-way bill generation for goods transport above Rs. 50,000

9. Security

InvoiceAI handles sensitive financial data and must maintain strict security standards:

- Invoice data encryption at rest (AES-256) and in transit (TLS 1.3)

- PCI-DSS compliance for payment link processing (no card data stored, handled by payment gateway)
- Comprehensive audit trail: every invoice creation, modification, payment, and reminder is logged with timestamp, actor, and IP address
- Role-based access control (RBAC): Owner, Admin, Accountant, and Viewer roles with granular permissions
- API rate limiting and API key rotation policies
- GSTIN validation against government database to prevent fraud
- Data retention policies aligned with Indian tax law (8 years for invoices and financial records)
- Regular security audits and penetration testing

10. Milestones

The InvoiceAI development follows an aggressive 10-week timeline to reach general availability:

Week	Phase	Deliverables
1-3	MVP	Invoice CRUD, GST calculation, PDF generation, email delivery, basic dashboard
4-5	Payments	Razorpay integration, payment tracking, basic bank reconciliation
6-7	AR Automation	WhatsApp integration, reminder sequences, AI message drafting
8	Intelligence	Payment prediction model, aging reports, cash flow forecasting
9-10	GA Launch	E-invoice integration, Tally export, production hardening, documentation

Post-GA roadmap includes mobile app, multi-language invoice support (Hindi, Tamil, Marathi), and AI-powered expense categorization.